

the three regions, well mixed, frontal or stratified, is generally the most productive; what are the relative contributions of the top mixed layer, thermocline and bottom mixed layer to the total production of the water column in the stratified regime; and how representative is a single station, like E1 (ref. 41), of production in the English Channel? The interrelationship between the distribution of phytoplankton and herbivorous zooplankton is poorly understood, particularly for the stratified regime. It seems probable that the high concentration of plant cells in the thermocline is an important food source for both migratory⁴² and non-migratory grazing organisms.

We thank Captain Dowell and the officers and crew of RV Sarsia for assistance at sea.

Received October 7; accepted November 18, 1975.

- ¹ Steele, J. H., *Nature*, **248**, 83 (1974).
- ² Platt, T., and Denman, K. L., *Mém. Soc. r. Sci. Liège*, **7**, 31 (1975).
- ³ O'Brien, J. J., and Wroblewski, J. S., *J. Theor. Biol.*, **38**, 197–202 (1973).
- ⁴ Platt, T., *Deep Sea Res.*, **19**, 183–188 (1972).
- ⁵ Denman, K. L., and Platt, T., *Mém. Soc. r. Sci. Liège*, **7**, 19 (1975).
- ⁶ Fasham, M. J. R., and Pugh, P. R., *Deep Sea Res.* (in the press).
- ⁷ Pingree, R. D., and Pennycuik, L., *J. mar. biol. Ass. U.K.*, **55**, 261–264 (1975).
- ⁸ Pingree, R. D., Pennycuik, L., and Battin, G., *J. mar. biol. Ass. U.K.*, **56**, 975–992 (1975).
- ⁹ Lumby, J. R., *Fishery Invest. Ser. 2*, **14**, 1–67 (1935).
- ¹⁰ Pingree, R. D., *J. mar. biol. Ass. U.K.*, **56**, 965–974 (1975).
- ¹¹ Matthews, D. J., *Rep. N. Sea Fish. Invest. Comm.*, **3**, 1906–08, 269–282 (1911).
- ¹² Pingree, R. D., *Deep Sea Res.*, **16**, 275–295 (1969).
- ¹³ Lorenzen, C. J., *Deep Sea Res.*, **13**, 223–227 (1966).
- ¹⁴ Grall, J. R., Fèvre-Lehoerff, G., and Fèvre, J. Le, *Cah. océanogr.*, **23**, 145–170 (1971).
- ¹⁵ Fearnhead, P. G., *Deep Sea Res.*, **22**, 311–321 (1975).
- ¹⁶ Simpson, J. H., and Hunter, J. R., *Nature*, **250**, 404–406 (1974).
- ¹⁷ Pingree, R. D., Forster, G. R., and Morrison, G. K., *J. mar. biol. Ass. U.K.*, **54**, 469–479 (1974).
- ¹⁸ Cooper, L. H. N., *J. mar. biol. Ass. U.K.*, **39**, 637–658 (1960).
- ¹⁹ Smayda, T. J., *A. Rev. Oceanography mar. Biol.*, **8**, 353–414 (1970).
- ²⁰ Eppley, R. W., Holm-Hansen, O., and Strickland, J. D. H., *J. Phycol.*, **4**, 333–340 (1968).
- ²¹ Stommel, H., and Fedorov, K. N., *Tellus*, **19**, 306–325 (1967).
- ²² Tait, R. I., and Howe, M. R., *Deep Sea Res.*, **15**, 275–280 (1968).
- ²³ Prandtl, L., *The Essentials of Fluid Dynamics* (Blackie, London and Glasgow, 1952).
- ²⁴ Pingree, R. D., and Griffiths, D. K., *Nature*, **250**, 720–722 (1974).
- ²⁵ Carslaw, H. S., and Jaeger, J. C., *Conduction of Heat in Solids* (Oxford University Press, Oxford, 1959).
- ²⁶ Channon, R. D., and Hamilton, D., *Nature*, **231**, 383–385 (1971).
- ²⁷ Eppley, R. W., *Fish. Bull. natn. oceanic atmos. Adm. U.S.*, **70**, 1063–1085 (1972).
- ²⁸ Wyatt, T., and Horwood, J., *Nature*, **244**, 238–240 (1973).
- ²⁹ Braarud, T., and Heimdal, B. R., *Nytt. Mag. Bot.*, **17**, 91–97 (1970).
- ³⁰ Hickel, W., Hagmeier, E., and Drebes, G., *Helgolander Wiss. Meeresunters.*, **22**, 401–416 (1971).
- ³¹ Ballantine, D., and Smith, F. M., *Br. phycol. J.*, **8**, 233–238 (1973).
- ³² Parsons, T. R., Stephens, K., and Strickland, J. D. H., *J. Fish. Res. Bd Can.*, **18**, 1001–1016 (1961).
- ³³ Holmes, R. W., Williams, P. M., and Eppley, R. W., *Limnol. Oceanogr.*, **12**, 503–512 (1967).
- ³⁴ Loftus, M. E., and Seliger, H. H., *Chesapeake Sci.*, **16**, 79–92 (1975).
- ³⁵ Bainbridge, R., *Biol. Rev.*, **32**, 91–115 (1957).
- ³⁶ Langmuir, I., *Science*, **87**, 119–123 (1938).
- ³⁷ Kamykowski, D., *J. mar. Res.*, **32**, 67–89 (1974).
- ³⁸ Fèvre, J. Le, and Grall, J. R., *J. exp. mar. Biol. Ecol.*, **4**, 287–306 (1970).
- ³⁹ Wilson, D. P., *Life of the Shore and Shallow Sea* (Nicholson and Watson, London 1935).
- ⁴⁰ Ryther, J. H., *Limnol. Oceanogr.*, **1**, 61–70 (1956).
- ⁴¹ Russell, F. S., Southward, A. J., Boalch, G. T., and Butler, E. I., *Nature*, **234**, 468–470 (1971).
- ⁴² Southward, A. J., *J. mar. biol. Ass. U.K.*, **42**, 275–375 (1962).
- ⁴³ Eppley, R. W., and Harrison, W. G., *AEC Progress Rep. Research on Marine food Chain*, 208–221 (1975).

Models for metal ion function in carbonic anhydrase

Paul Woolley

Max-Planck Institute for Biophysical Chemistry, 34 Göttingen-Nikolausberg, West Germany

A model catalyst is described which has properties in common with carbonic anhydrase. The model demonstrates the availability of a mechanism, previously only hypothetical, for the action of the enzyme. It also shows, however, that this mechanism alone is not adequate to produce the high activity of the enzyme.

DETECTED in 1928, discovered in 1932, recognised to contain zinc in 1939, obtained pure in 1959, crystallised in 1962 and the subject of some 1,800 papers and articles, carbonic anhydrase continues to elude a full mechanistic description, even though this enzyme is small, stable and easily isolated, and its X-ray structure is known to a resolution^{1,2} of 2 Å.

Carbonic anhydrase catalyses a range of reactions, including the hydration of carbon dioxide and aldehydes, and the hydrolysis of carboxylic, sulphonic and carbonic esters. The CO₂ hydration is apparently the reaction of biological importance, both in respiration and in intracellular CO₂–HCO₃[–] equilibrium. In most of these reactions the rate in the absence of enzyme is believed to be limited by the nucleophilic attack of an oxygen atom or ion at an electrophilic centre, usually carbon, illustrated in Fig. 1a. The enzyme accelerates the CO₂ hydration, the acetaldehyde hydration and the hydrolysis of *p*-nitrophenyl acetate by factors of 10⁹, 10⁵ and 10⁵ respectively over the reaction in water.

The zinc ion which occurs naturally in carbonic anhydrase may be replaced artificially by other metal ions, but only cobalt(II) and zinc confer appreciable activity on the enzyme; with no metal or with other metals the enzyme is practically inactive in hydration.

The enzyme possesses an ionising group with a *pK_a* of

about 7; only the alkaline form of the enzyme is catalytically active.

One elegant mechanistic suggestion³ is that a metal-bound water molecule could ionise to provide the necessary nucleophile in the form of metal-bound hydroxide (Fig. 1b). Other possibilities include the ionisation of a nearby histidine group to provide a general-base catalyst at the active site⁴; the zinc could then act as a point of substrate binding, perhaps also polarising and thereby activating the substrate by virtue of its Lewis acid strength⁵. Whatever the mechanism of nucleophilic attack, other stages of the reaction may also be catalysed, such as proton transfer within the enzyme–substrate complex⁶ or between the enzyme and its aqueous environment⁷.

The purpose of this article is to re-examine the plausibility of the first mechanism referred to above, the zinc-bound hydroxide nucleophile (ZHN) hypothesis. Reviews and background literature are given in refs 2 and 8.

A priori arguments against the ZHN mechanism have been twofold: first, that a zinc-bound water cannot have such a low *pK_a* as 7 (refs 4 and 9); second, that a zinc-bound hydroxide ion would be so polarised by the zinc as to lose most of its negative charge to the zinc and thus be too weak a nucleophile to attack, for example CO₂ (refs 6 and 10 and Fig. 1b). Transition-metal hydroxy complexes can indeed attack CO₂ (refs 11 and 12), but so far few quantitative data are available.

To investigate the universality of these two presuppositions, model systems were chosen to reflect closely the desired properties of the enzyme. The much-abused term “model” is here understood as a simple chemical system with a property or properties relating it to the biological system under consideration. Ideally a range of model compounds, structurally controlled to reproduce in varying degrees the property in question, provide a quantitative scale on which the biological system

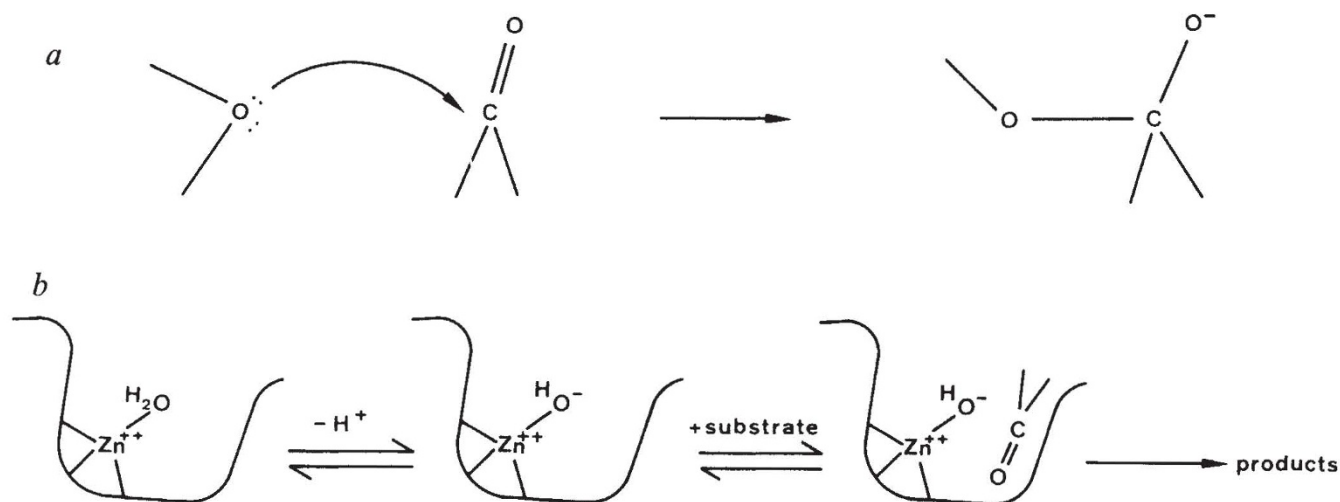


Fig. 1 Nucleophilic attack. *a*, Negative charge on oxygen enables it to react through its outer electron pairs with a positive centre such as the carbonyl carbon shown here; *b*, a postulated mechanism for nucleophilic attack in carbonic anhydrase.

might fall; if it continues to deviate from this scale then it has still not been fully understood. A model system may alternatively be designed to reproduce one of the biological properties of the system, for instance, to test the hypothesis that a sufficient condition for the action of an enzyme is a particular conjunction

of reacting atoms or groups; if successful this does not confirm that the biological system acts in the same way as the model, but it does show that the action of the simple system is in principle available for nature to make use of in the complex one.

An ionised water molecule?

Discussion of factors influencing the ionisability of water bound to zinc is bedevilled by the possible formation of hydrolysed cluster species and by the uncertain coordination number of the metal ion when water molecules are counted^{13,14}, as well as by the complexity of the equilibria present^{14,15}. To overcome these problems, macrocyclic ligands of the type CR (ref. 16 and refs therein) (Fig. 2) were used. Here the four nitrogen donor atoms are perforce bound to the metal, giving room for one or two water molecules. Electronic spectroscopy¹⁸ and X-ray crystallography¹⁷ support coordination numbers in aqueous solution of five for CoCR²⁺ and ZnCR²⁺, and six for NiCR²⁺ and CuCR²⁺.

Titration of the complexes in aqueous solution with sodium hydroxide reveals the pattern of behaviour shown in Table 1; two points emerge. First, the metals which can activate carbonic anhydrase can also promote the ionisation of water in their CR complexes. If this is due to the reduced coordination number of zinc or cobalt then the same ionisation should be possible in carbonic anhydrase. Second, nickel and copper do not promote this ionisation, although they are normally stronger Lewis acids, forming stronger complexes, than zinc or cobalt (II) (for example ref 18). A tempting conclusion is that whatever chemical influence determines the ionisation pattern of the CR complexes must also operate in carbonic anhydrase. This is, however, a *non sequitur*; a more logical approach is required.

The stronger acidity of five-coordinate ZnCR²⁺ and CoCR²⁺ than that of their corresponding hexaquo ions is probably determined by their reduced coordination number, as can be argued on simple electrostatic grounds^{13,19}, reducing the overall coordination number increases the total positive charge on zinc and the resulting polarisation of the bound H₂O. A further reduction in *pK_a* when the coordination number is reduced to four, as found in carbonic anhydrase¹, would therefore not be surprising. So far, carbonic anhydrase is the only zinc complex known for certain to have a coordination sphere composed of three nitrogens plus one oxygen in solution, although there are other candidates²⁰ for this. Similar titrimetric investigations using acyclic polyamine ligands point to the same correlation between coordination number and *pK_a*, (refs 21 and 22) with the additional insight that a less polar environment for the complex encourages ionisation still further,

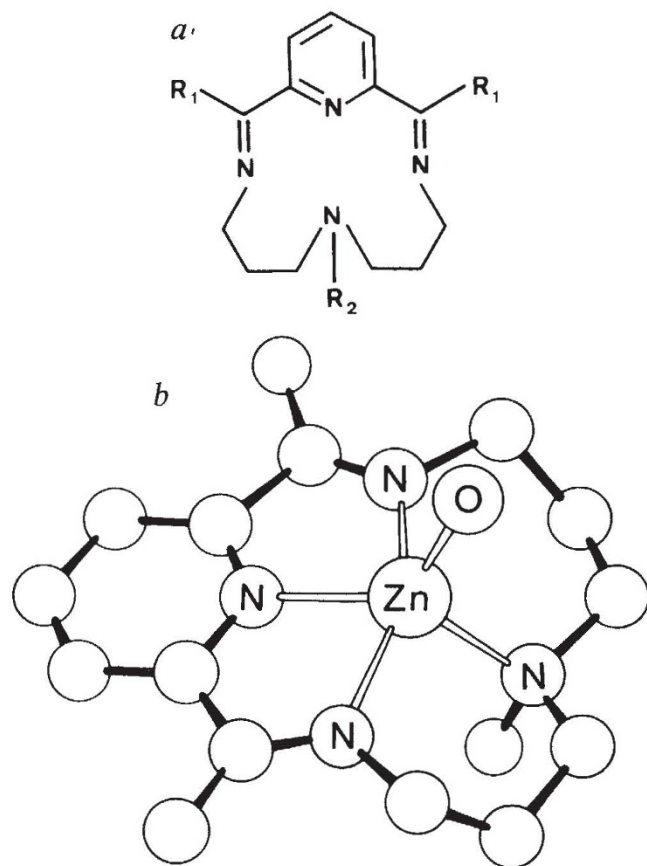


Fig. 2 Macrocyclic ligands. *a*, Free ligand. *b*, In the complex the four nitrogen atoms bind a metal ion—Zn²⁺, Cu²⁺, Ni²⁺ or Co²⁺—which also binds water; the skeleton of the Zn(N-MeCR)(H₂O)²⁺ complex is shown (H omitted, unlabelled atoms are C.) CR: R₁=CH₃, R₂=H; N-MeCR: R₁=R₂=CH₃; desdiMeCR: R₁=R₂=H.

perhaps to the extent of 1–2 units, as plainly shown by methylation of the NH_2 groups of the polyamine²³. This factor could well operate at the relatively non-polar active site of the enzyme. The provision of a firm binding site for zinc, of low coordination number and low polarity, and accessible to substrate, could indeed be the reason for the surprisingly large size of the protein in comparison with the substrate.

The probable higher coordination of the copper and nickel CR complexes similarly explains their lesser polarising effect on bound water molecules. Such a coordination number would reflect the known lesser propensity of zinc and cobalt, in comparison with copper and nickel, towards the formation of geometrically regular complexes, and indeed this very property, which imparts a greater coordination flexibility to Zn^{2+} and Co^{2+} , has been speculatively put forward as determinative of the activity of carbonic anhydrase with zinc and cobalt²⁴. The active site of carbonic anhydrase appears to be

poor a nucleophile to have a catalytic role in carbonic anhydrase. In addition, it was felt desirable to see what sort of modifications short of complexation by apocarbonic anhydrase would induce catalytic activity of any kind in zinc ions. The aim was thus to find a catalyst containing zinc ions along with organic groups and/or the hydroxide ion.

Carbonic anhydrase's range of substrates is wide, so it was necessary at the start to restrict the range of reactions examined. One reaction was studied intensively—the hydration of acetaldehyde. This was chosen as it is very efficiently catalysed by carbonic anhydrase and for reasons of experimental convenience.

The basic mechanism of acetaldehyde hydration has been well reviewed²⁵.

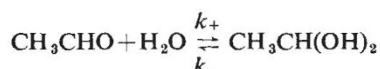


Table 1 Ionisation of water bound to metals in macrocyclic complexes, in aqueous solution at low ionic strength (0.002–0.04 mol l⁻¹)

Ligand	M	M(ligand)OH ₂ ²⁺ Temperature (°C)	K_a $\rightleftharpoons$	M(ligand)OH ⁺ + H ⁺ pK_a^*	Approximate heat of ionisation (kcalorie mol ⁻¹)
CR	Zn	25		8.69†	} 7.1±1
CR	Zn	0		9.17	
CR	Cu	25		> 11	
CR	Ni	25		> 11	
CR	Co	25		8‡	} 8.3±1
N-MeCR	Zn	25		8.12	
N-MeCR	Zn	0		8.68	
N-MeCR	Cu	25		> 11	
desdiMeCR	Zn	25		8.13	} 6.0±1
desdiMeCR	Zn	0		8.53	
(H ₂ O) ₆	Zn,Co	—		~10	
Apocarbonic anhydrase	Zn,Co	—		~7	

* $pK_a = -\log_{10} K_a$. Error estimated from several determinations < 0.05 units.

† This supercedes the value given in ref. 19.

‡ Estimated value; apparent cluster formation prevented a more accurate determination.

fairly accommodating to the geometry preferred by the metal (ref. 25 and ref. 2, 642–643). Following this argument, one could say: if the activity in carbonic anhydrase is determined by an ionising metal-bound water molecule, then the non-activity of Cu^{2+} and Ni^{2+} in carbonic anhydrase could be a consequence of the non-ionisability or non-existence of this water, the latter a consequence of the coordination number taken up by Cu^{2+} or Ni^{2+} as the resultant of the different geometries preferred by the metal and the apoenzyme respectively. This must however be regarded as an hypothesis to be investigated and not as an explanation, particularly in view of the existence of an (as yet unidentified) ionising group in the inactive Mn^{2+} , (ref. 46) Cu^{2+} (E. Grell and A. McConnell, personal communication) and $(\text{VO})^{2+}$ (ref. 27) substituted enzymes.

The heat of ionisation of bovine carbonic anhydrase, 7 kcalorie mol⁻¹, has been taken as an indication that the ionising group is histidine, whose imidazole group has a similar heat of ionisation²⁸. The data in Table 1 shows that this value is equally compatible with the ionisation of zinc-bound water. The point of emphasis here, however, is the potential stability of a zinc-bound hydroxide ion, irrespective of whether it is formed by dissociation of a water molecule or, for example, by the addition of hydroxide accompanying a conformational change of the protein.

A zinc-bound hydroxide nucleophile?

The purpose of the kinetic studies here was to test the second presupposition above—that metal-bound hydroxide is too

In contrast to carbonic anhydrase, aqueous zinc ions are only poor catalysts for the reaction, but there is a very simple, general catalytic effect in which zinc ions and other species in solution work together to catalyse the hydration reaction³⁰. These other species include in particular hydroxide ions, and this effect was investigated further in the hope that it might also throw some light on the ZHN hypothesis. Unfortunately, the cooperation between zinc and hydroxide ions is less amenable to systematic mechanistic study, as practical difficulties are caused by polynucleation and precipitation. To study this reaction further it was therefore necessary to use a different system containing zinc and hydroxide, and the natural choice was the ZnCR^{2+} complex, which was already available and whose properties were well investigated.

The efficiency of a catalyst in acetaldehyde hydration is defined as $dk_+/d[\text{catalyst}]$. Experimental details of the procedure used to measure the rates are given elsewhere (ref. 30 and refs therein). At low pH the efficiency of ZnCR^{2+} is low, but at high pH it becomes very high (Fig. 3). This behaviour superficially resembles that of carbonic anhydrase³¹.

Mechanism of ZnCR^{2+} catalysis

The efficiencies of various catalysts are shown in Table 2. Empirically, ZnCR^{2+} is a poor catalyst (Fig. 3, low pH) and ZnCR.OH^+ , the form of the complex predominating at high pH, a good one (Fig. 3, high pH). Thus the ZnCR^{2+} and OH^- cooperate to produce an enhanced catalysis, just as hoped for. A detailed analysis of the kinetics and activation parameters

(to be published) shows that the hydration proceeds through collision between ZnCR.OH^+ and CH_3CHO . Two possible mechanisms for this are shown in Fig. 4. Borate is a more powerful base than ZnCR.OH^+ , yet its catalytic effect is much smaller (Table 2). Therefore ZnCR.OH^+ cannot simply be acting as a general base (Fig. 4a), as then borate would be the better catalyst; it must be attacking the carbonyl group directly (Fig. 4b). Space-filling models suggest that simultaneous binding to zinc of hydroxide and aldehyde is forbidden by the geometry of the CR ligand; this point is being further investigated.

Table 2 Catalytic efficiency in acetaldehyde hydration

Catalyst	Efficiency	Reference
CH_3COO^-	0.023	30
CH_3COOH	0.073	
Zn^{2+}	0.0084	
$\text{H}_2\text{O}^{\text{aq}}$	0.00014	
ZnCR^{2+}	0.036	
CuCR^{2+} (pH 5.0, 7.5)	0	This work
CuCR^{2+} (pH 9.5)	2 ± 1 (?)	
ZnCR.OH^+	196	
Zn(N-MeCR).OH^+	112	
$\text{Zn(desdiMeCR).OH}^+$	130	
OH^-	58,000	32
Carbonic anhydrase	14*	31
Borate	$\lesssim 7$	This work

Units are $\text{mol}^{-1} \text{l s}^{-1}$.

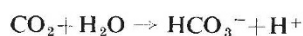
*The turnover number (700 s^{-1}) of the enzyme is divided by 50 mol l^{-1} , to obtain an order-of-magnitude correction for the presumed high concentration of catalyst at the active site.

To obtain a comparison, the catalytic properties CuCR^{2+} , NiCR^{2+} and CoCR^{2+} were examined in the same way. Unfortunately the nickel and cobalt complexes reacted directly with the acetaldehyde (shown by ultraviolet spectroscopy) and so no conclusion could be reached in these cases. CuCR^{2+} was unaffected by the acetaldehyde and showed no catalytic properties at pH values of 5.0, 7.5, or 9.5. This is consistent with the absence of a CuCR.OH^+ species in this pH range, demonstrated by titration (Table 1).

That the catalysis in the presence of CR involves direct attack of zinc-bound hydroxide backs up the hypothesis that the same attack occurs in the above-mentioned cooperative catalysis by zinc and hydroxide ions.

Carbon dioxide hydration

The availability of zinc-bound hydroxide as a good nucleophile provides an obvious next step for investigation of its catalytic ability in other reactions, particularly those catalysed by carbonic anhydrase. Most interesting would be the hydration of CO_2 , for which carbonic anhydrase's catalytic power is the greatest. This was examined qualitatively by mixing saturated CO_2 solutions with buffered solutions of ZnCR^{2+} at 0°C . The initial rate of pH change was measured and converted by comparison with an HCl titration to the rate of liberation of protons; this gave the reaction rate, for the stoichiometry is



at the pH used (9.2). The empirical catalytic constant of the ZnCR.OH^+ complex is given by

$$\frac{1}{[\text{CO}_2]} \cdot \frac{d[\text{CO}_2]}{dt} \cdot \frac{1}{[\text{ZnCR.OH}^+]} \quad (3)$$

and was determined as about $25 \text{ mol}^{-1} \text{l s}^{-1}$. Since previous work has demonstrated the absence of general-base catalysis in the hydration of CO_2 (refs 33 and 34) the most obvious

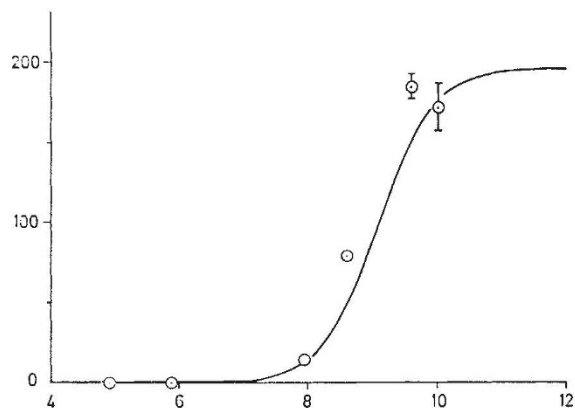


Fig. 3 The catalytic efficiency of ZnCR^{2+} in the hydration of acetaldehyde, as a function of pH. The solid line is the curve calculated assuming that the active species is ZnCR.OH^+ with an efficiency of $196 \text{ mol}^{-1} \text{l s}^{-1}$. The pK_a found is 9.07 (compare the value from titration, 9.17, Table 1). Increasing errors at high pH made measurement above pH 10 impossible.

mechanism available for attack is analogous to that of Fig. 4b. ZnCR.OH^+ is quite a good catalyst (Table 4), although still not comparable with carbonic anhydrase.

A similar result has been reported¹² using the complex $\{\text{Cu.glycylglycinate.OH}\}$, in which the active species was likewise considered to be a metal-bound hydroxide ion. ZnCR.OH^+ is a less ambiguous catalyst to interpret, as no potentially active ligand groups other than OH^- are present, and the metal has no further free coordination sites.

Relationship between model and enzyme

The applicability of the enzyme model to the enzyme can now be considered in relation to the ZHN hypothesis.

The titrimetric studies on ZnCR^{2+} and related complexes, and similar studies on a range of polyamines (refs 22, 23 and unpublished), make it clear that the ionisation of zinc-bound water is to be expected, on condition that the coordination number of zinc be reduced from six to five or four—a condition fulfilled by carbonic anhydrase. This does not prove that the ZHN in carbonic anhydrase exists, but it means, should a zinc-bound water in carbonic anhydrase be shown not to ionise, that this in itself would be a remarkable property of the enzyme, and in need of special explanation.

Table 3 Catalytic efficiency in carbon dioxide hydration

Catalyst	Efficiency	Reference
H_2AsO_3^-	0.9	33*
HAsO_4^-	0.002	
BrO^-	2	
SeO_3^-	1.2	
SO_3^-	0.2	
NO_2^- , NO_3^- , CO_3^{2-} , SO_4^{2-}	0	34*
Complexes MX_n , where $\text{M} = \text{Mn, Fe, Co, Ni, Cu, Zn}^{2+}$ and $\text{X} = \text{EDTA, nitrilotriacetic acid, dipyr-ridyl, cysteine, mercaptoethylamine and cyanide}$	0	
$\text{Cu(glycylglycinate).OH}$	20	
ZnCR.OH^+	25	
Carbonic anhydrase	$10,000^\dagger$	
		35

Units are $\text{mol}^{-1} \text{l s}^{-1}$.

*In these references different units are used. The data have been converted so as to accord with the definition given in this article.

[†]See footnote to Table 2.

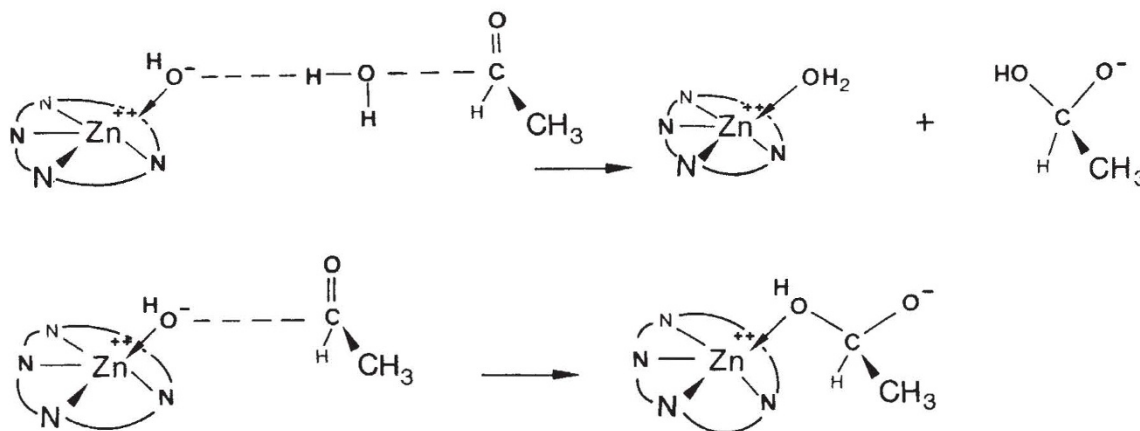


Fig. 4 a, General-base and b, direct (nucleophilic) attack by ZnCR.OH^+ . Only the rate-determining step is shown; proton transfer and ligand replacement at zinc are assumed to be fast.

The kinetic work has shown that a zinc-bound hydroxide ion can act as a very powerful nucleophile, even though its proton affinity is about 10^7 times that of a free hydroxide ion. Its nucleophilicity is great enough to account for the enzyme's catalysis of acetaldehyde hydration, though not of carbon dioxide hydration (Tables 2 and 3). The existence of this nucleophile in the enzyme and in the model does not mean that the mechanisms in the enzyme and in the model are the same; indeed, the rates for CO_2 are so different that, even if there is a ZHN in carbonic anhydrase, the enzyme must still have an extra means of assisting the hydration which the acetaldehyde, because of its different structure, cannot utilise. There is also evidence^{36,37} that an additional, tightly-bound bicarbonate ion could have such a role by mediating proton transfer in the zinc-substrate complex.

The evidence for the ZHN postulate on the basis of enzyme studies is far from overwhelming; model studies cannot add to it directly, although they appear to increase the previous plausibility and the comprehensibility of such a mechanism. What happens when one tries to extend the hypothesis to the metal-substituted carbonic anhydrases, with the thought that their activity or inactivity could be related to the presence or absence of an ionising water molecule at the metal? At first sight this seems reasonable. The flexible ions Zn^{2+} and Co^{2+} accept the distorted geometry and low coordination number offered by the enzyme, while copper and nickel do not, thus failing to polarise the bound water sufficiently to induce ionisation. Manganese is coordinatively flexible, but polarises bound water only weakly (it forms in general comparatively weak complexes¹⁸, and in carbonic anhydrase seems to be less capable than zinc of polarising sulphonamide inhibitors³⁸).

Before such a theory could be accepted, however, several obstacles would need to be removed. (1) Ionising groups are associated with metal in Mn^{2+} , Cu^{2+} and $(\text{VO})^{2+}$ -substituted carbonic anhydrases (see above). These would have to be shown not to be metal-bound water. (2) In the acid form of cobalt-carbonic anhydrase the metal ion is apparently very poorly accessible to exchanging water molecules³⁹. Also, the spectroscopic change associated with the activity-linked ionisation in the cobalt enzyme is very marked⁹, whereas the spectroscopic effects of water and hydroxide ligands are almost identical⁴⁰, and the spectrum of CoCR.OH_2^{2+} is correspondingly similar to that of CoCR.OH^+ (unpublished). Thus, even if the ZHN hypothesis is correct, certain features of the metal's coordination geometry require clarification. (3) A kinetic effect remains unexplained. The forms of zinc-carbonic anhydrase with the more acidic proposed zinc-bound water molecules—having presumably the most strongly polarising zinc ions—should by virtue of the diminished negative charge on the hydroxide's oxygen atom possess the weakest nucleophiles, with the smallest rate of attack on substrates. The reverse order is seen, how-

ever^{41,44}. This is also the case with the simple compounds $\text{ZnOH}^+_{\text{aq}}$ and ZnCR.OH^+ (Table 2) for the substrate acetaldehyde. Nucleophilicity and basicity are not always closely related⁴², but a satisfactory explanation of this would still be welcome.

Conclusions from models

Models show that the presence of the postulated zinc-bound hydroxide ion in carbonic anhydrase at neutral pH values would be in accord with reasonable expectation. This hydroxide ion would have sufficient nucleophilic power to account for the enzyme's activity in acetaldehyde hydration, although not in carbon dioxide hydration. The ZHN postulate for the enzyme led to the observation of surprisingly high nucleophilicity in the zinc-bound hydroxide. Thus the uses and limitations of chemically modelling a biological system are illustrated.

I thank many colleagues for discussion and criticism and in particular Dr R. H. Prince for encouragement and support. Financial means were provided by the SRC, Magdalene College and the Alexander von Humboldt Foundation. Most of the experimental work was carried out at the University Chemical Laboratories, Cambridge.

Received July 25; accepted October 30, 1975.

- Kannan, K. K., et al., *Proc. natn. Acad. Sci. U.S.A.*, **72**, 51–55 (1975).
- Lindskog, S., et al., in *The Enzymes*, (edit. by Boyer, P. D.) 587–665 (Academic, New York).
- Davis, R. P., in *The Enzymes*, second ed., 5 (edit. by Boyer, P. D., Lardy, H., and Myrback, K.), 545–562 (Academic, New York).
- Pocker, Y., and Meany, J. E., *Biochemistry*, **6**, 668–678 (1967).
- Williams, R. J. P., *Inorg. Chim. Acta Revs.*, **5**, 75–82 (1971).
- Wang, J. H., *Proc. natn. Acad. Sci. U.S.A.*, **66**, 874–881 (1970).
- Cayley, G. R., thesis, Univ. Kent (1973).
- Silverman, D. N., and Tu, C. K., *J. Am. Chem. Soc.*, **97**, 2263–2269 (1975).
- Coleman, J. E., in *Inorganic Biochemistry* (edit. by Eichhorn, E.), 488–548 (Elsevier, Amsterdam, 1973).
- Lindskog, S., *J. biol. Chem.*, **238**, 945–951 (1963).
- Pesando, J. M., *Biochemistry*, **14**, 681–688 (1975).
- Flynn, B. R., and Vaska, L., *J. Am. Chem. Soc.*, **95**, 5081–5083 (1973).
- Breslow, E., in *The Biochemistry of Copper* (edit. by Peisach, J., Aisen, P., and Blumberg, W. E.), 149–156 (Academic, New York, 1966).
- Cayley, G. R., thesis, Univ. Kent (1973).
- Appleton, D. W., and Sarkar, B., *Proc. natn. Acad. Sci. U.S.A.*, **71**, 1686–1690 (1974).
- Martin, R. B., *Proc. natn. Acad. Sci. U.S.A.*, **73**, 4346–4347 (1974).
- Prince, R. H., Stotter, D. A., and Woolley, P. R., *Inorg. Chim. Acta*, **9**, 51–54 (1974).
- Caira, M. R., Nassimbeni, L. R., and Woolley, P. R., *Acta Crystallogr.*, **B31**, 1334–1338 (1975); Caira, M. R., Nassimbeni, L. R., Wolfe, M. A., and Woolley, P. R., *Inorg. Chem.* (in the press).
- Cotton, F. A., and Wilkinson, G., *Advanced Inorganic Chemistry*, third ed., 588–597 (Interscience, New York, 1972).
- Prince, R. H., and Woolley, P. R., *Nature phys. Sci.*, **240**, 117–118 (1972).
- Paoletti, P., Fabbri, L., and Barbucci, R., *Inorg. Chim. Acta Revs.*, **7**, 43–68 (1973).
- Barbucci, R., Fabbri, L., Paoletti, P., and Vacca, A., *J. Chem. Soc. Dalton Trans.*, 1763–1767 (1973).
- Jonassen, H. B., and Thielemann, H., *Z. anorg. allgem. Chem.*, **320**, 274–282 (1963).
- Coates, J. H., Gentle, G. J., and Lincoln, S. F., *Nature*, **249**, 733–775 (1974).
- Prince, R. H., and Woolley, P. R., *Angew. Chem.*, **84**, 461–471 (1972); *Angew. Chem. Int. Ed.*, **11**, 408–417 (1972).
- Malkin, R., and Malmström, B. G., *Adv. Enzymol.*, **33**, 177–244 (1970).
- Laird, A., Gradsztajn, S., and Navon, G., *FEBS Lett.*, **30**, 351–355 (1973).
- Fitzgerald, J. J., and Chasteen, N. D., *Biochemistry*, **13**, 4338–4347 (1975).
- Pocker, Y., and Stone, J. T., *Biochemistry*, **7**, 4139–4145 (1968).
- Bell, R. P., *Adv. phys. org. Chem.*, **4**, 1–29 (1966).

- ³⁰ Woolley, P. R., *J. Chem. Soc., Dalton Trans.*, 1570–1575 (1975).
³¹ Pocker, Y., and Meany, J. E., *Biochemistry*, **4**, 2535–2541 (1965).
³² Bell, R. P., Rand, M. H., and Wynne-Jones, K. M. A., *Trans. Farad. Soc.*, **52**, 1093–1102 (1956).
³³ Dennard, A. E., and Williams, R. J. P., *J. chem. Soc., A*, 812–816 (1966).
³⁴ Dennard, A. E., thesis Univ. Oxford (1963).
³⁵ Kernohan, J. C., *Biochim. biophys. Acta*, **96**, 304–317 (1965).
³⁶ Gerster, R. H., Maren, T. H., and Silverman, D. N., *Proc. int. Conf. Stable Isotop. Chem. Biol. Med.*, first, (edit. by Klein, P. D.), 219–228 (1973).
³⁷ Yeagle, P. L., Lochmüller, C. H., and Henkens, R. W., *Proc. natn. Acad. Sci. U.S.A.*, **72**, 454–458 (1975).
³⁸ Lanir, A., and Navon, G., *Biochemistry*, **11**, 3536–3544 (1972).
³⁹ Fabry, M. E., Koenig, S. H., and Schillinger, W. E., *J. biol. Chem.*, **245**, 4256–4262 (1970).
⁴⁰ Cotton, F. A., Wilkinson, G., *Advanced Inorganic Chemistry*, third ed., 577 (Interscience, New York, 1972).
⁴¹ Ward, R. L., *Biochemistry*, **9**, 2447–2454 (1970).
⁴² Basolo, F., and Pearson, R. G., *Mechanisms of Inorganic Reactions*, second ed. 396–403 (Wiley, New York, 1967).

Conservation of cytoplasmic poly(A)-containing RNA in mouse and rat

M. Rosbash

Rosenstiel Basic Medical Sciences Research Center, Brandeis University, Waltham, Massachusetts 02154

M. S. Campo

Department of Genetics, University of Edinburgh

K. S. Gummerson

Medical Research Council Mammalian Genome Unit, Department of Zoology, University of Edinburgh, West Mains Road, Edinburgh EH9 3JT, UK

By comparing the melting temperature of DNA-DNA duplexes between related species, it is shown that total single-copy DNA evolves at a faster rate than DNA which is transcribed into poly(A)-containing RNA. Such a comparison suggests that at least 70% of rat single-copy DNA does not code for protein.

THE organisation and function of the DNA in the eukaryotic genome are the subject of much speculation and research^{1,2}. There exists a variety of models which concern themselves with the regulation of gene activity. At present, however, few data show what fraction of eukaryotic DNA is genetically active, that is, what fraction of the DNA codes for protein. An alternative way of posing the same question is: how many different proteins are synthesised throughout the life of the organism?

Various estimates of this number make distinctly different predictions. Based largely on genetic analyses of *Drosophila* lethal mutations, the minimal estimate gives

approximately 5,000 genes encoded by the *Drosophila* genome³. Mammalian haploid genomes contain 3 pg of DNA⁴. Approximately 70% of the genome is single-copy DNA, known to be the site of transcription of most of the messenger RNA. Assuming a number-average molecular weight of 6×10^5 for eukaryotic mRNA⁵, 5,000 different mRNAs (the minimal estimate for *Drosophila*) represent only 0.5% of the single-copy fraction of mammalian DNA. If the number of genes is proportional to the genome size, this estimate for mammals increases to 50,000 different mRNA species and 5.0% of the available single-copy DNA. From a consideration of the mutation rate observed for amino acid changes in protein, Kimura has concluded that the cost to a species due to deleterious mutations would be very high if all the DNA of eukaryotes were expressed genetically⁶, in accordance with these low estimates of the protein coding fraction of the genome. Alternatively, there may be many more than 50,000 different mRNA species and, therefore, much more than 5% of the single-copy DNA of mammals may be protein-coding DNA. In the extreme,

Table 1 Extents of reaction and thermal stabilities of homologous and heterologous hybrids with rat ³H-cDNA and rat ³²P-scDNA

Experiment	Homologous						Heterologous					
	Reassociation temperature (°C)	Rat DNA % Rxn.	scDNA % Rxn.	ΔT_m (°C)	cDNA % Rxn.	ΔT_m (°C)	Mouse DNA % Rxn.	scDNA % Rxn.	ΔT_m (°C)	cDNA % Rxn.	ΔT_m (°C)	
a	60	90	86	0.5	87	3.0	90	67	14.0	82	8.0	
b	70	85	83	0.5	72.5	3.0	78	31	9.0	65	6.0	
c	73	75	73	0.5	67	2.5	74	18.5	8.0	51	6.0	
d	75	78	77.5	0.5	70	3.0	72	9.5	6.0	35.5	6.0	
e	75	—	—	—	—	—	—	1.3	—	4.0	—	
f	75	—	—	—	—	—	—	1.6	—	—	—	
SI Nuclease												
Experiment	Homologous						Heterologous					
	Reassociation temperature (°C)	scDNA % Rxn.	T_m (°C)	cDNA % Rxn.	T_m (°C)		scDNA % Rxn.	T_m (°C)	cDNA % Rxn.	T_m (°C)		
g	75	56	66.5	65	66.5		7	64.5	37	64.5		

The data for experiments a, b, d, and g are taken from Figs 1, 2, 3, and 4, respectively. Experiment c is identical to experiments a, b, and d with reassociation temperature of 73 °C. In experiments e and f, radioactive tracer DNA was incubated in the absence of non-radioactive driver DNA as in Fig. 1.